THE UNIVERSITY OF SYDNEY

COMP9120: Semester 2 2026

Campus Pantry

FoodHub booking, attendance and inventory database

Ziyue Wu, Nikhil Khatri, Samowel Aboud
Group 94

The problem

Food collections need reliable reservations, enough staff and clear stock records. Separate records make it harder to connect demand, attendance and inventory.

A linked database supports collection planning and helps staff investigate stock discrepancies.

The solution

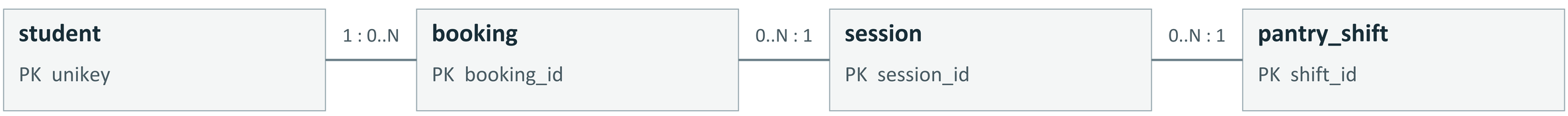
A PostgreSQL showcase for students, volunteers and staff. Students reserve collection sessions, staff record attendance, and inventory records connect purchases to physical counts.

16 tables **3 reporting views** **12 query groups**
Daily setup: 3 shifts and 12 half-hour sessions.

Python command-line interface — Parameterised psycopg2 queries ——— **PostgreSQL**

Database design

Selected relations from the 16-table schema. Cardinalities show permitted records.



booking references student and session; session references pantry_shift.



Each batch references a purchase line; each physical count references a batch.

Shared identity and roles

person stores identity once. student, volunteer and staff share its key and may overlap. Assignment tables link staff and volunteers to shifts.

Separation and trade-offs

product stores product details. shift_stock_count and stock_adjustment support reconciliation. stock_batch retains product_id, with a composite foreign key enforcing its purchase-line match.

Booking transaction

1 Lock the student record.
2 Lock session; validate time and staff override.
3 Request an insert with a random token.
4 Trigger rechecks capacity and daily limits.
5 Commit on success; roll back on error.

At most 50 non-cancelled bookings per session. One active daily booking by default. An extra daily booking needs staff identification and a reason; capacity still applies.

Showcase workflows

Collection and attendance

Staff enter tokens for same-day attendance, 10:00-16:00 Sydney. Late arrivals remain valid after their shift closes.

Staffing and shift lifecycle

Staff approve volunteer requests. Staffed shifts follow PLANNED, OPEN and CLOSED.

Inventory operations

Record a real opening before receipts, waste or closing. Close reconciled shifts to protect stock records.

A stock decision

Opening + receipts - waste - closing

100 + 10 - 2 - 88 = 20 rice packs

Verified synthetic showcase. Dispensing is an estimate, not a count of individual handouts.

Validation and scope

44/44 tests passed twice on PostgreSQL 16.15, including 10 concurrency scenarios. UniKey sign-in and token entry are simulated. QR images, email delivery and university authentication remain outside scope.

Acknowledgements

Based on Group 94 project code and documents. OpenAI Codex assisted with source review, implementation, tests, documentation and layout.
Sources: revised 01_schema.sql, services.py and evidence/test-results-run-1.json and run-2.json. Synthetic data only.